

GOAL:

Decide which marketing strategy works the best using A/B testing

TARGET METRIC:

promotion	location_count	mean_of_aggregated_sales	std_of_aggregated_sales
1	43	58.1	16.03
2	47	47.33	14.5
3	47	55.36	16.38

The stricter significance level ($\alpha = 0.01$) was used to account for multiple pairwise comparisons and reduce the risk of Type I error.

HYPOTHESIS TESTING:

Promotion 1 vs Promotion 2

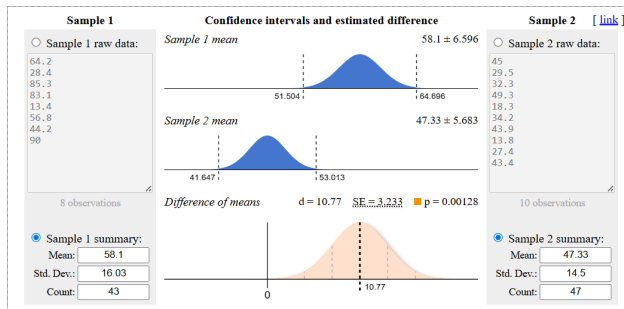
H0: There is no difference in average sales between promotion 1 and Promotion 2

H1: There a difference in average sales between promotion 1 and Promotion 2

Question: Does the average value differ across two groups?

Conclusion:	Reject H_0
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p= 0.00128	<	0.01
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Verdict: Sample 1 mean is greater

Hypothesis: ☒ $d = 0$ ☐ $d \leq 0$ ☐ $d \geq 0$

Confidence: 99%

Promotion 1 vs Promotion 3

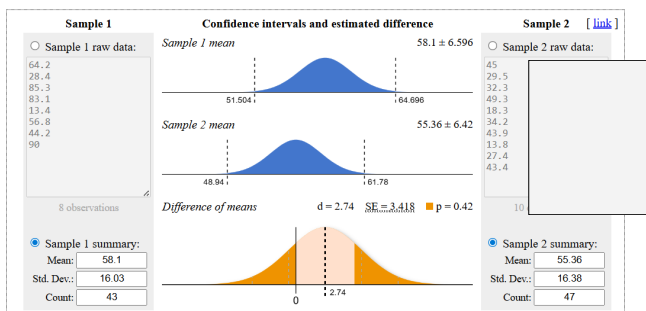
H0: There is no difference in average sales between promotion 1 and Promotion 3

H1: There a difference in average sales between promotion 1 and Promotion 3

Question: Does the average value differ across two groups?

Conclusion:	Fail to reject H_0
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p= 0.42	>	0.01
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Verdict: No significant difference

Hypothesis: ☒ $d = 0$ ☐ $d \leq 0$ ☐ $d \geq 0$

Confidence: 99%

Promotion 2 vs Promotion 3

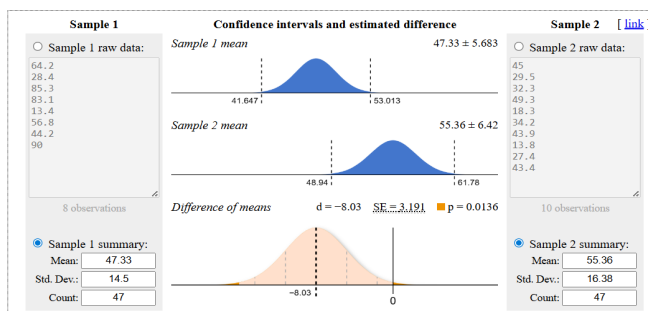
H0: There is no difference in average sales between promotion 2 and Promotion 3

H1: There a difference in average sales between promotion 2 and Promotion 3

Question: Does the average value differ across two groups?

Conclusion:	Fail to reject H_0
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p=0.0136	$\geq$	0.01
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Verdict: No significant difference

Hypothesis: ☒ $d = 0$ ☐ $d < 0$ ☐ $d > 0$

Confidence:

Key Findings

GOAL

Fast-food company planning to add new menu items needs to determine which of 3 promotions delivers the highest return through A/B testing.

HYPOTHESIS

Since we have to compare multiple promotions the Hypothesis is:

H0: There is no difference in average sales between promotions

H1: There is a difference in average sales between promotions

METHODOLOGY

T-testing was performed with a significance level of $\alpha = 0.01$ (99% confidence)

to account for multiple pairwise comparisons and reduce the risk of Type I error.

TESTING RESULTS

Comparison	p value vs $\alpha = 0.01$	Outcome	Verdict
Promotion 1 vs Promotion 2	0.00128 < 0.01	Reject H0	Significant difference
Promotion 1 vs Promotion 3	0.42 > 0.01	Fail to reject H0	No Significant difference
Promotion 2 vs Promotion 3	0.0136 >= 0.01	Fail to reject H0	No Significant difference

SUPPORTING ANALYSIS

Additional data exploration confirmed Promotion 1's strength across all segments:

- Consistently highest average sales by market size

- Strong performance across all weeks (1-4)

- Outperforms in all store age groups

RECOMMENDATION

Promotion 1 is the recommended strategy. It significantly outperforms Promotion 2 ($p = 0.00128$) and shows consistent strength across all market segments.

Although no statistically significant difference was found between Promotion 1 and Promotion 3 at the 99% confidence level,

Promotion 1 shows consistently higher average performance across all analyses.

IMPORTANT NOTE

The stricter $\alpha = 0.01$ minimizes false positives (Type I error) but increases the risk of missing real differences (Type II error).

Follow-up testing between Promotion 1 and 3 with larger sample sizes may be beneficial for definitive confirmation if needed.

LIMITATIONS

Comparison	Sample size (n)	Margin of Error	Confidence Interval	Power
Promotion 1 vs Promotion 2	n1= 43 n2= 47	8.50	2.51 to 19.03	Sufficient for large effect
Promotion 1 vs Promotion 3	n1= 43 n3= 47	8.99	-6.26 to 11.74	Small effect, low power
Promotion 2 vs Promotion 3	n2= 47 n3= 47	8.39	-16.42 to 0.36	Insufficient at 99% level

t (from t-table):

2.63

Promotion 1 vs Promotion 3

Shows a small effect size and a wide confidence interval (-6.26 to 11.74), which includes zero.risk of a Type II error.

This indicates high uncertainty, and the test likely lacks sufficient power to detect small effects.

Suggestion:

Since the observed effect size between Promotion 1 and Promotion 3 is small and the confidence interval is wide, no clear conclusion can be drawn.

It is recommended to collect additional data or extend the duration of the experiment to increase statistical power.

Promotion 2 vs Promotion 3

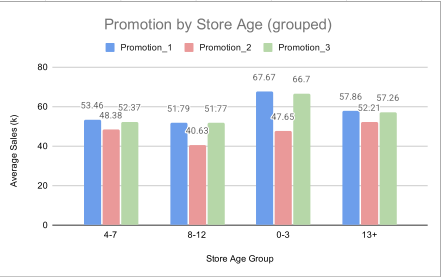
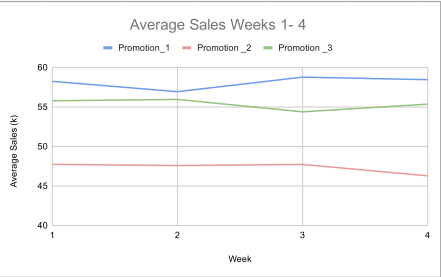
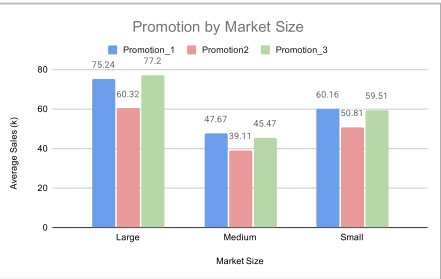
Although a moderate effect size is observed, the confidence interval (-16.42 to 0.36) includes zero, indicating that the result is not statistically significant at the 99% level.

This suggests insufficient power to reliably detect moderate effects, and the conclusion remains uncertain.

Additional data would be required to confirm whether a true difference exists.

Suggestion:

It is recommended to increase the sample size and re-run the test to improve statistical power.



1	SELECT	Goal	Explored unique values and first insight of how many promotions are in dataset
2	COUNT(DISTINCT location_id) as unique_locations,		
3	COUNT(DISTINCT promotion) as unique_promotions,		
4	COUNT(*) as total_records		
5	FROM `turing_data_analytics.wa_marketing_campaign`;		
6			
Link to query: https://console.cloud.google.com/bigquery?sq=147855269776:eafb1c7975804f32962ba5f77bbe54e			
3	SELECT	Goal:	Aggregate by LocationID and PromotionID before conducting the statistical tests
4	location_id,		
5	promotion,		
6	COUNT(*) as record_count,		
7	ROUND(AVG(sales_in_thousands), 2) as avg_sales,		
8	ROUND(SUM(sales_in_thousands), 2) as total_sales,		
9	ROUND(STDDEV(sales_in_thousands), 2) as std_dev,		
10	FROM `turing_data_analytics.wa_marketing_campaign`		
11	GROUP BY location_id, promotion		
12	ORDER BY location_id, promotion		
1	WITH	Goal:	Get overview statistics
2	aggregated AS(		
3	SELECT		
4	location_id,		
5	promotion,		
6	COUNT(*) as record_count,		
7	ROUND(AVG(sales_in_thousands), 2) as avg_sales,		
8	ROUND(SUM(sales_in_thousands), 2) as total_sales,		
9	ROUND(STDDEV(sales_in_thousands), 2) as std_dev,		
10	FROM `turing_data_analytics.wa_marketing_campaign`		
11	GROUP BY location_id, promotion		
12	ORDER BY location_id, promotion		
13	)		
14			
15	SELECT		
16	promotion,		
17	COUNT(*) as location_count,		
18	ROUND(AVG(avg_sales), 2) as mean_of_aggregated_sales,		
19	ROUND(STDDEV(avg_sales), 2) as std_of_aggregated_sales,		
20	FROM aggregated		
21	GROUP BY promotion		
22	ORDER BY promotion;		
23			
25	/*	Goal:	Data Validation to make sure that the pools are good for comparison
26	-- Validation: Check data points for statistical testing.		
27	-- What to look for: Balanced groups (similar number of locations per promotion)		
28	SELECT		
29	promotion,		
30	COUNT(*) as num_locations_tested,		
31	ROUND(AVG(record_count), 2) as avg_records_per_location,		
32	SUM(record_count) as total_original_records		
33	FROM aggregated		
34	GROUP BY promotion;		
35	*/		
36			
37	/*		
38	--Validation: Verify locations are evenly distributed		
39	--What to look for: Empty or very few rows		
40	SELECT		
41	location_id,		
42	promotion,		
43	COUNT(*) as promotion_count		
44	FROM aggregated		
45	GROUP BY location_id, promotion		
46	HAVING COUNT(*) > 1;		
47	*/		
1	SELECT *	Goal:	Get data on how promotions perform on average WoW
2	FROM (		
3	SELECT		
4	week,		
5	promotion,		
6	ROUND(AVG(sales_in_thousands), 2) AS avg_sales		
7	FROM `turing_data_analytics.wa_marketing_campaign`		
8	GROUP BY week, promotion		
9	)		
10	ORDER BY week;]		
1	SELECT	Goal:	Get data on how promotions perform accross market sizes
2	promotion,		
3	market_size,		
4	ROUND(AVG(sales_in_thousands), 2) AS avg_sales		
5	FROM `turing_data_analytics.wa_marketing_campaign`		
6	GROUP BY promotion, market_size		
7	ORDER BY market_size, promotion;		
1	SELECT	Goal:	Get data on how promotions perform accross store age groups
2	promotion,		
3	CASE		
4	WHEN age_of_store <= 3 THEN '0-3'		
5	WHEN age_of_store <= 7 THEN '4-7'		
6	WHEN age_of_store <= 12 THEN '8-12'		
7	ELSE '13+'		
8	END AS age_group,		
9	ROUND(AVG(sales_in_thousands), 2) AS avg_sales		
10	FROM `turing_data_analytics.wa_marketing_campaign`		
11	GROUP BY promotion, age_group		
12	ORDER BY age_group, promotion;		

How promotions perform on average week on week				Market Size			Age group		
week	_1	_2	_3	promotion	market_size	avg_sales	promotion	age_group	avg_sales
1	58.24	47.73	55.78	1	Large	75.24	1	0-3	67.67
2	56.93	47.58	55.95	2	Large	60.32	2	0-3	47.65
3	58.77	47.72	54.38	3	Large	77.2	3	0-3	66.7
4	58.45	46.28	55.35	1	Medium	47.67	1	13+	57.86
				2	Medium	39.11	2	13+	52.21
				3	Medium	45.47	3	13+	57.26
				1	Small	60.16	1	4-7	53.46
				2	Small	50.81	2	4-7	48.38
				3	Small	59.51	3	4-7	52.37
							1	8-12	51.79
							2	8-12	40.63
							3	8-12	51.77